

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



232/3

Physics (Practical)

Paper

3

Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education

INSTRUCTIONS TO CANDIDATES:-

- Write your **name**, **indexnumber** and **school** in the spaces provided above.
- Answer **all** the questions in the spaces provided in the question paper
- You are supposed to spend the first **15 minutes** of 2 ½ hours reading the whole paper carefully before commencing your work.
- Marks are given for a clear record of the observations actually made, their suitability, accuracy and the use of them
- Record your observations as soon as you make them.
- Mathematical tables, slide rules and silent non-programmable electronic calculators may be used.
- Take $g = 10\text{ms}^{-2}$

FOR EXAMINERS' USE ONLY:

QUESTION	MAXIMUM SCORE	CANDIDATE SCORE
1	20	
2	20	
TOTAL SCORE	40	

For Marking Schemes Contact 0746-222-000 / 0742-999-000

1. PART A

You are provided with the following:

- a voltmeter ($0 - 5V$)
- a carbon resistor labelled R
- Two new size D dry cells
- cells holder
- a capacitor labelled C ($2200\mu F$)
- a stopwatch
- eight connecting wires at least four of which have crocodile clips
- a switch

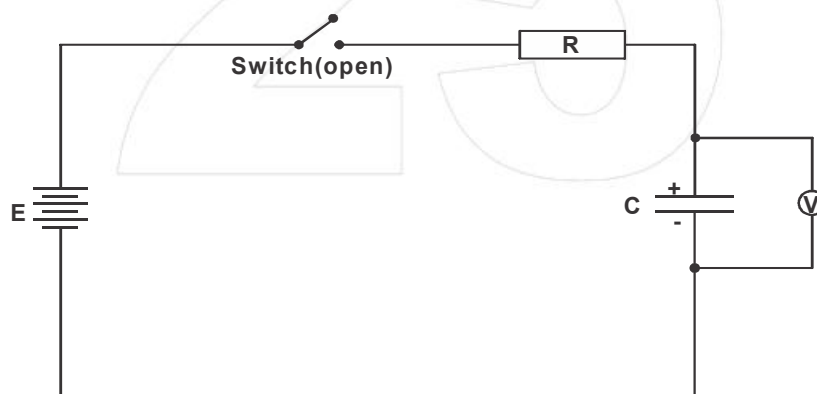
Proceed as follows :

- i) a) Draw a circuit diagram having a switch, voltmeter, two dry cells and connecting wires that can be used to determine the Emf of the cells provided. **(1mk)**

- b) Use the above circuit to determine the Emf of the cells

Emf = V **(1mk)**

- ii) Connect the circuit as shown in figure 1 below.



- iii) Close the switch and note the maximum voltage attained on the voltmeter. Record it as $V_{\max}$

$V_{\max}$ = V **(1 mark)**

iv) Open the switch and simultaneously start your stopwatch and note the time taken for the voltmeter reading to decrease from $V_{\max}$ to 0.2V. record this value in table 1 below. Short the terminals of the capacitor C to attain a reading of zero volts on the voltmeter.

v) Repeat procedure iii and iv for the other values of the voltage indicated in table 1

Table 1

Voltage, V (volts)	0.2	0.5	0.8	1.0	1.5	1.8
Time (t)(s) taken for voltage to decrease from $V_{\max}$ to V						

vi) determine the resistance R of the resistor given that $t = 70/11CR$. Where t is the time at which $V=0.5V$ and C is the capacitance of the capacitor.

3mks

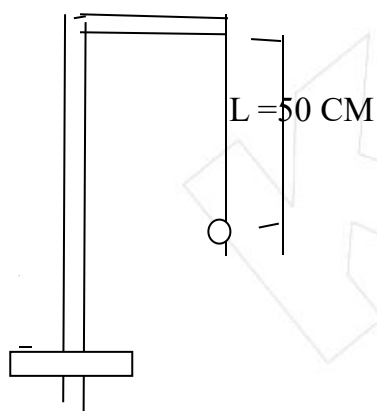
PART B

You are provided with:

- *A complete stand*
- *A stop watch*
- *A light string*
- *A pendulum bob or a 50g mass*
- *A metre rule or a half metre rule*

Proceed as follows:

i) Set up the apparatus as shown in the figure below



ii) Displace the bob slightly at a small angle from the vertical line and record the time taken 20 oscillations in the table below. Repeat the for two more trials

Table 2

1 ½ mks

Trial 0	Trial 1	Trial 2	Trial 3
Time for 20 oscillations			

iii) Find the average time for 20 oscillations

1 ½ mks

iv) Calculate the average periodic time T_{av}

v) Given that $T_{av} = 2\pi\sqrt{l/g}$ find the value of g

3mks

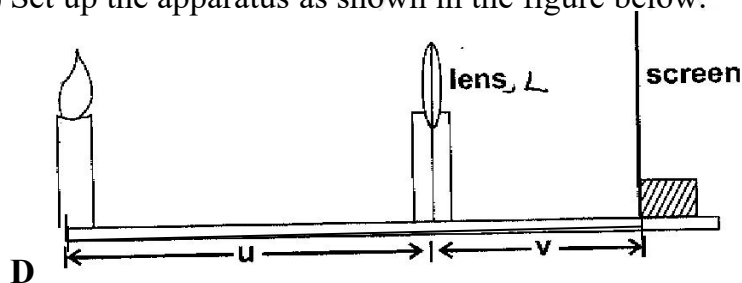
Question 2

You are provided with the following:-

- A biconvex lens.
- A lens holder.
- A screen.
- A meter rule.
- A candle
- A cross wire

Proceed as follows:

(a) Set up the apparatus as shown in the figure below.



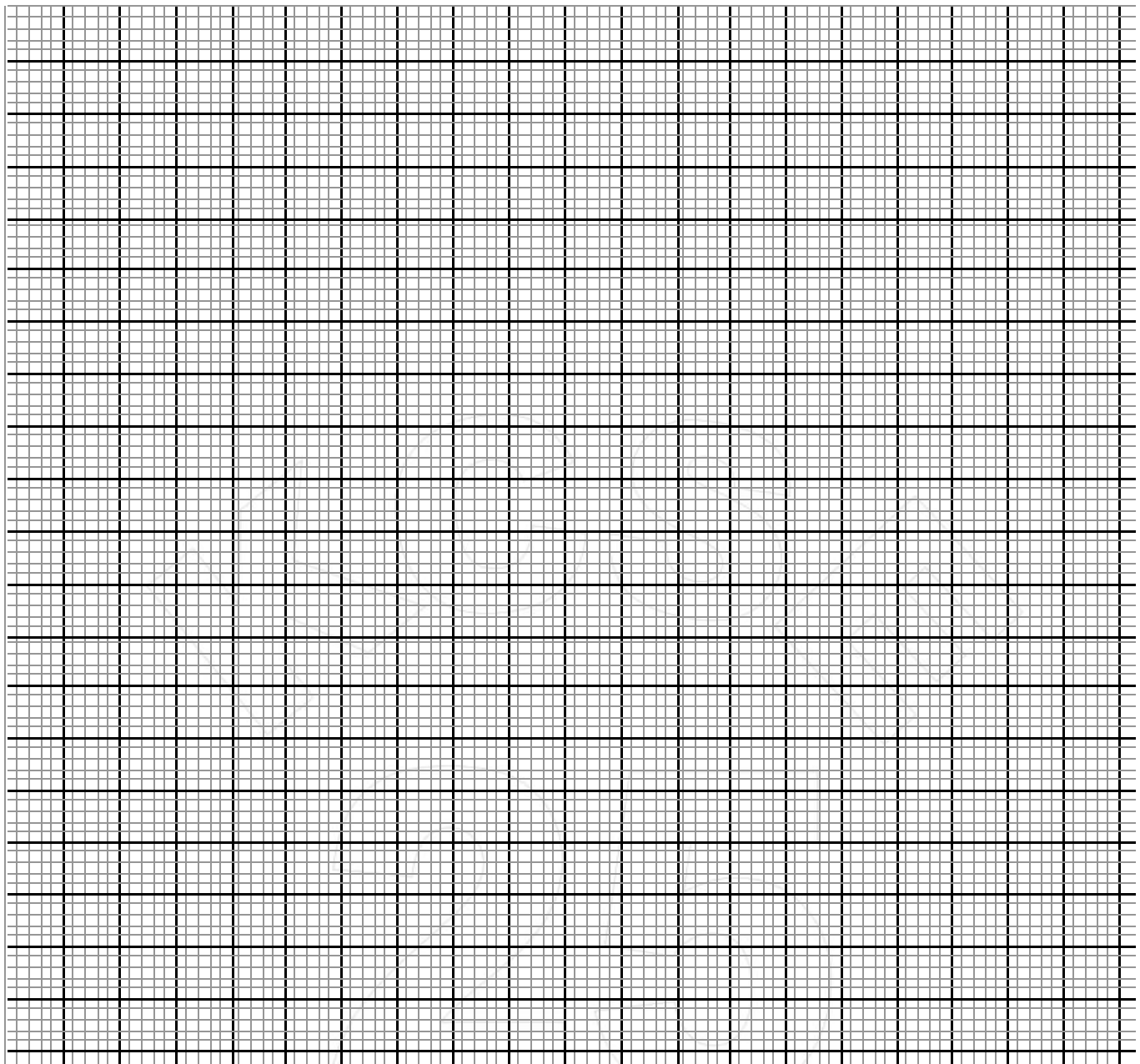
NB: THE CROSS WIRE PROVIDED MUST BE PLACED JUST INFRONT OF THE CANDLE TO ACT AS THE OBJECT

- (b) Place a lit candle and a cross wire at a distance $U = 20\text{cm}$ from the lens
- (c) Adjust the position of the screen until a sharp image of the cross wire is focused on the screen.
Record d in the table 2.
- (d) Repeat part (b) and (c) for other values shown in table 2 and complete the table. (8mks)

Table 2

U(cm)	20	25	30	35	40	45
D (cm)						
V=D-U(cm)						
$1/m = u/v$						

- (e) Plot a graph of $1/m$ against U . (5mks)



f) determine the slope S of the graph

(3mks)

g) given that $1/f = 1/v + 1/u$, use the graph to determine the focal length of the lens **4mks**

